

Many abstract def<sup>n</sup>s today! "centipede math"

But focus on  $\mathbb{R}^n$ , its subspaces, matrices

↙ Enough for now! We'll see that for finite dimensional

Vector spaces, we can do everything by working in  $\mathbb{R}^n$ .

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Show in any given v-space  $V$ :

0 vector is unique, additive inverse is unique,  $c \cdot \vec{0} = \vec{0}$ ,  $0 \cdot \vec{v} = \vec{0}$ ,  $(-1) \cdot \vec{v} = -\vec{v}$

I A vector space over  $\mathbb{R}$  consists of the following data  $\leftarrow \text{Def}^n_5$

- a set  $V$  whose elts are called vectors
- a map  $V \times V \xrightarrow{+} V$  called addition (denoted  $+$ ) such that
  - (1)  $+$  is associative i.e.  $v+(w+u) = (v+w)+u$
  - (2) there is a neutral vector called " $0$ " i.e.  $0+v=v=v+0$
  - (3) there is a map  $V \rightarrow V$  giving "inverses" to  $+$  i.e.  $v+(-v)=0=(-v)+v$
  - (4)  $+$  is commutative i.e.  $v+w=w+v$
- a map  $\mathbb{R} \times V \rightarrow V$  called scalar multiplication such that

$$c \cdot (v+w) = (c \cdot v) + (c \cdot w) \quad (b+c) \cdot v = b \cdot v + c \cdot v \quad (bc) \cdot v = b \cdot (c \cdot v) \quad 1 \cdot v = v \quad \text{for } \begin{cases} b, c \in \mathbb{R} \\ v, w \in V \end{cases}$$

Rmk:  $V$  can't be empty b/c  $0 \in V$

II A map  $f: V \rightarrow W$  between Vect. spaces is called linear if  
 $f(v_1+v_2) = f(v_1) + f(v_2)$   
 $f(c \cdot v_1) = c \cdot f(v_1)$

①-④: def of abelian gp

①-③: def of group



## Examples

①  $\mathbb{R}^n$  [with  $+$ ,  $\cdot$  defined slotwise]

② set of all  $r \times c$  matrices ( $r, c$  fixed  $\in \mathbb{Z}_{>0}$ )

③  $\mathbb{R}^{\omega} =$  set of all sequences of real nos.

④  $\{ax^2+bx+c \mid a, b, c \in \mathbb{R}\}$  add, scale as polynomials

III A subspace  $W$  of a v.sp.  $V$  is a subset of  $V$  that is a vector space (in its own right) under the same operations as in  $V$ . (For this it is enough for

$W$  to be closed under  $+$  and  $\cdot$  nonempty.)

• Fix a set  $S$ .  $\{f: S \rightarrow \mathbb{R}\}$  is a vector space with  $+$  &  $\cdot$  pointwise

Examples of subspaces

①  $S = \{1, \dots, n\} = [n]$  ③  $S = \mathbb{N}$

②  $S = [r] \times [c]$

Given a linear map  $f: V \rightarrow W$  we have

$\ker(f) = \{v \mid f(v) = 0\}$  is a subspace of  $V$

$\text{im}(f) = \{f(v) \mid v \in V\} \subseteq W$

$\mathbb{R}^{\omega} \supset \mathbb{R}^{\infty}$   
eventually 0 sequences



Words that carry over for a given subset  $S \subset V$ -space  $V$

- linear combination of vectors in  $S$
- $\text{span}(S)$  is a subspace of  $V$ .

•  $S$  is linearly dependent if some nontrivial lin. comb. of elts of  $S$  is  $0$  i.e.  
 $c_1 v_1 + \dots + c_n v_n = 0$  for some  $\{v_i \in S\}$   
 $\{c_i \in \mathbb{R}, \text{ some } c_i \neq 0\}$ .

•  $S$  is linear independent if it is not linearly dependent.

• A basis for  $V$  is a linearly indep subset  $B \subset V$  s.t.  $\text{span}(B) = V$ .

( $B$  is a basis for  $V \iff$  each vector in  $V$  is a unique linear combination of vectors in  $B$ )

•  $V$  is called finite dimensional if there is a finite set that spans  $V$ .

Claim: Any f.dim  $V$ -space  $V$  has a basis and any two bases have the same finite cardinality. [Pf: later]